

Question Number	Answer	Additional guidance	Mark
8(a)	An explanation that makes reference to three of the following points: • fewer young tree shoots eaten (after wolves were reintroduced) (1) • (because) elk would not graze undisturbed when wolves were around (1) • more (young tree shoots) were eaten in forest habitat (than riverside habitat) / fewer (young tree shoots) were eaten in riverside habitat (than forest habitat) (1) • (because) the mature trees provided some {camouflage / protection} (from the wolves) (1)	ACCEPT the percentage (of young tree shoots eaten) decreased ACCEPT converse ACCEPT because some elk have been eaten by wolves / fewer elk because wolves are predators (of elk) ACCEPT {larger/more rapid} decrease (in number of shoots eaten) in riverside habitat ACCEPT (because) elk can hide (from the wolves) in the forest / elk have no {hiding places / protection} from being seen (by the wolves) by the river ACCEPT predation of elk is higher in riverside (habitat)	(3)

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8(b)	A description that makes reference to the following points: • as the number of elk (per km²) increases the average beaver lifespan decreases (1) • {larger/rapid} decrease {initially / as population increases from 30 elk (per km²)} (1) • very little change in average lifespan of beavers when the number of elk increase above {70/80/90} elk per km² (1)	ACCEPT negative correlation ACCEPT {exponential / non-linear} decrease ACCEPT the beaver lifespan plateaus when the number of elk increase above {70/80/90} elk per km²	(3)

Question Number	Answer	Additional guidance	Mark
8(c)(i)	An answer which includes the following points: • (count) number of species (1) • {area / size} of habitat (1)	ACCEPT per unit area	(2)

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8(c)(ii)	• $(D =) \frac{N(N-1)}{\sum n(n-1)}$	ACCEPT other correct equations e.g. simple biodiversity index = species richness ÷ species evenness	(1)

Question Number	Answer
*8(d)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. <u>Basic</u> • increasing numbers of wolves (until 2003) / decrease in number of wolves (from 2007 to 2010) • overall increase in wolves / 20 more wolves in 2010 (than in 1995) • the numbers of elk decreased as they were hunted by the wolves • overall decrease in numbers of elk (from 1995 to 2010) • the number of cottonwood shoots rapidly increased (after 2003) • the number of beaver colonies increased from 1 (in 1999) to 12 (in 2009) <u>Linkage</u> • the numbers of cottonwood shoots increased {when the numbers of elk decreased / as they weren't being eaten} • beavers and elk have similar niches / beavers were outcompeted for young tree shoots by elk • beaver {numbers/colonies} started to increase when {elk population size decreased / more shoots were available} • discussion of causes of decrease in wolf population e.g. decrease in elk food source • introduction of wolves increases biodiversity <u>Sustained</u> • {more ponds / new habitat} increase {number of species / species richness / biodiversity} • more trees (woodland and riverside habitats) increase {number of species / species richness / biodiversity} • biodiversity increases due to additional species in Yellowstone National Park e.g. wolf (6) Expert

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Information from one section
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Information from two sections linkage of 2 concepts
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Level 2 plus Discussion of how biodiversity would be increased